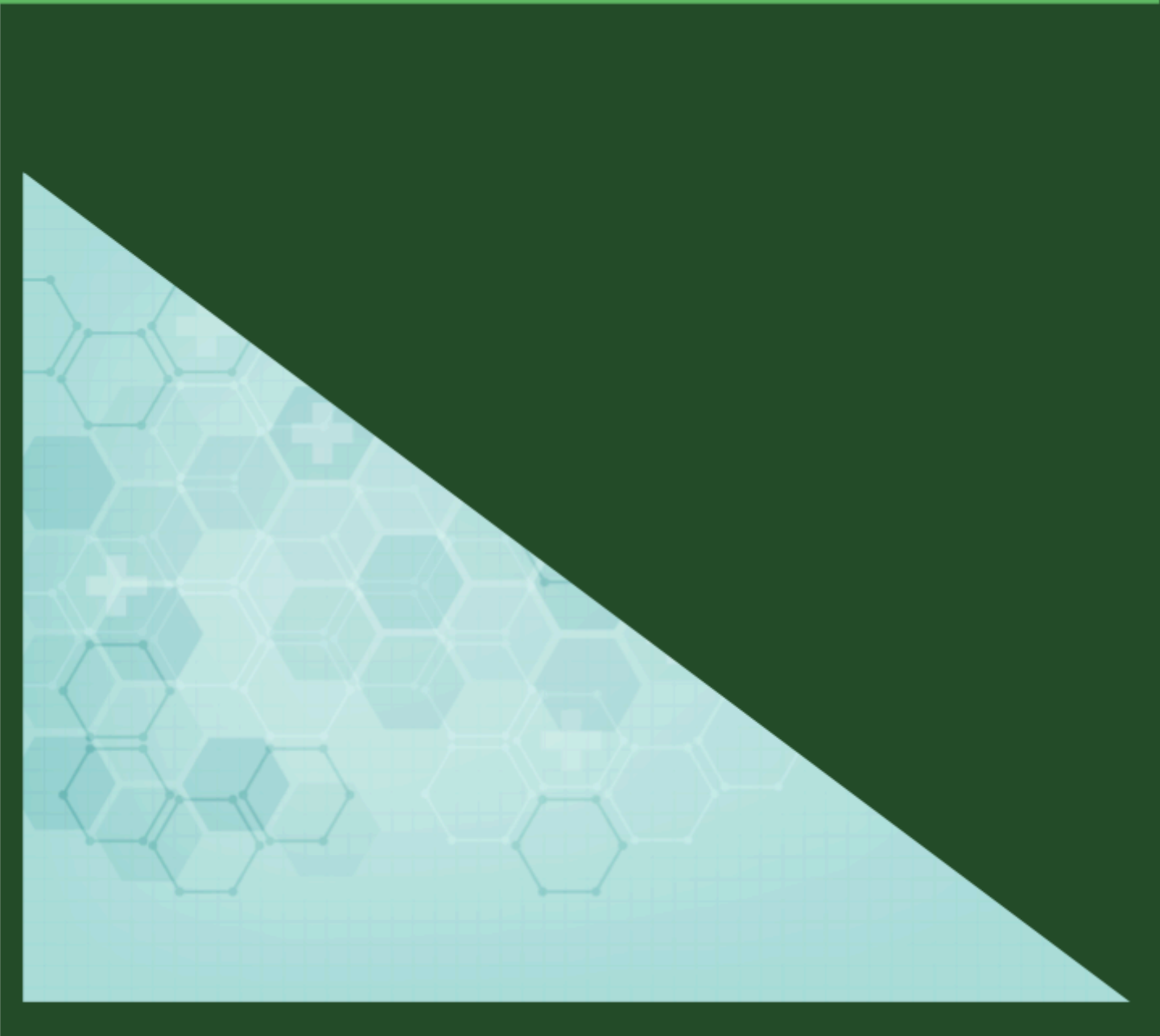




Climate Action Award

The Cambridge High School

GreenVolt - Trash Auto Sorting and Heat System

BUSINESS PLAN



مؤسسة الإمارات
EMIRATES FOUNDATION





Please use this template Business Plan to support your pitch, the plan will be shared with judges ahead of the pitch day final.

Send to FL@emiratesfoundation.ae

Deadline: 12pm 19th February 2025

Planning is vital to ensure that you build a solid business foundation for your project, proving to judges that you have a sound business plan that has explored all key areas for bringing your idea to life.

Make sure you include enough information to provide the judges confidence, and a full understating of your projected plans and objectives.

-
- The process of thinking and writing the plan provides clarity to your idea
 - If capital/resources are needed from beyond the AED200,000 prize fund, judges want to see a plan that demonstrates a solid understanding and vision for your idea
 - The plan will help you to prioritise and demonstrate the thinking behind the first year tasks that are most important
 - With growth, the plan offers a common understanding of the vision and predicted impact on positive climate behaviours and outcomes
 - This plan will provide your team something you should continually review and update over time
-

Write the plan keeping in mind that it is for the judge's reference. It needs to be understandable, readable, achievable and realistic.

This template is organised into 4 sub-sections to be completed:

1. Executive Summary
2. Project Definition/Overview
3. Operational Plan/Roadmap/ Financials
4. Assessment and Monitoring Plan

It is recommended to complete the Executive Summary last once all the other sections have been completed. As you move through each section, the writing and information should tell the story of your motivation, your vision, why you will be successful, how you will achieve success and how you will measure it.

Be very clear on the positive impact of your proposal, including social, community and physical benefits.



EXECUTIVE SUMMARY

Executive Summary

Opportunity:

Schools face significant challenges in waste management and energy consumption, with high volumes of waste ending up in landfills and excessive energy use for heating and cooling. GreenVolt addresses both of these issues by integrating cutting-edge AI, waste-to-energy systems, and sustainable technologies to reduce waste, save energy, and engage students in environmental stewardship.

Mission:

GreenVolt's mission is to revolutionize waste management and energy use within schools. By utilizing smart technology and waste-to-energy processes, we aim to reduce the environmental impact of school operations, lower energy costs, and foster a culture of sustainability among students, staff, and the broader community.

Our Solution:

Our project is not just about waste management—it's about redefining the role of auto-sorting bins as a system that actively recovers energy and resources. Over the 12-month action plan, our focus progresses from efficient waste sorting to integrating heat recovery and alternative energy extraction from non-recyclable waste. By harvesting waste heat and optimizing material separation for secondary resource recovery, we are transforming what would have been discarded into a valuable input for sustainability. This makes it a project that evolves over time, not just managing waste but redefining its purpose. This shift represents a game-changing approach, demonstrating that even discarded materials can be repurposed in meaningful ways.

GreenVolt combines AI-powered waste sorting, on-site composting, and energy recovery to significantly reduce school waste, lower energy consumption, and create educational opportunities. This solution is unique in that it integrates both waste management and energy generation, setting a new standard for sustainability within educational settings.

Focus:

This project is primarily targeted towards students, staff, and the school community, with a particular emphasis on fostering student leadership in sustainability initiatives. It also seeks to engage families and local businesses by encouraging sustainable practices beyond the school setting, creating a wider impact on the community and environment.

Expected Positive Outcomes:

In the first year, we expect to achieve several key milestones:

- **Months 1-3:** Implement AI-powered auto-sorting bins and sensors for better waste tracking.
- **Months 4-6:** Introduce composting units and thermal storage tanks to start capturing and storing thermal energy from organic waste.
- **Months 7-9:** Utilize the stored thermal energy for applications like water heating and classroom temperature regulation.
- **Months 10-12:** Convert thermal energy into electricity to power lighting and cooling systems.

These actions will result in a significant reduction in waste sent to landfills, a decrease in energy consumption, and an increase in sustainability awareness among students and staff.



GreenVolt's unique approach, combining waste management and energy recovery, will not only optimize school operations but also foster long-term positive environmental and social change. With scalability in mind, the project is designed to expand to other schools and communities, creating a ripple effect of sustainability across the region.

PROJECT OVERVIEW

● Mission Statement:

GreenVolt is committed to transforming waste into a valuable energy source through smart waste management and heat recovery systems. Our mission is to **reduce landfill waste, cut carbon emissions, and create a sustainable energy model** that enhances school operations while educating students about climate action. By integrating **AI-driven waste sorting, composting, and thermoelectric energy conversion**, we aim to make sustainability a practical, accessible, and impactful part of everyday school life.

● Team Narrative:

Our team has both the school's STEM Lead and Career Lead, along with students who are passionate about sustainability and technology. The idea for autosorting bins came from a common issue we noticed—recyclable waste often ends up in the wrong bin, making recycling efforts less effective. We wanted to create a solution that eliminates sorting errors while also encouraging better waste management habits in our school.

Last year, we participated in the Global Innovation Challenge and placed in the top three. The judges saw potential in our idea and encouraged us to take it further, which motivated us to refine and expand our concept. Sustainability has always been important to us, not just in school but also at home. Some of our parents work in ADNOC's sustainability sector, giving us insight into real-world waste management challenges and the importance of resource efficiency. Inspired by their expertise and our own drive to make a difference, we set out to design an intelligent waste-sorting system that makes recycling easier and more effective for everyone.

● Opportunity:

Schools face a dual challenge: **inefficient waste management**—resulting in contamination, low recycling rates, and excessive landfill contributions—and **high energy consumption**, particularly for heating and cooling. **GreenVolt presents an innovative, circular solution** that transforms waste into an asset, simultaneously tackling both issues by:



- **Enhancing waste segregation** with AI-powered smart bins, ensuring optimal recycling and waste processing.
- **Minimizing organic waste** through on-site composting, closing the loop on food and biodegradable waste.
- **Harnessing waste heat** to offset energy demands, converting thermal energy into a usable resource.
- **Immersing students in hands-on sustainability practices**, integrating AI, automation, and environmental science into real-world applications.

Who Stands to Gain?

Students: Engage with cutting-edge sustainability solutions, gaining firsthand experience in AI-driven waste management and energy recovery.

Schools: Lower operational costs through smarter waste management and reduced energy expenses, while advancing their sustainability goals.

The Environment: Achieves a **30-40% reduction in landfill waste**, curtails emissions from waste decomposition, and fosters a culture of responsible consumption and resource efficiency.

By seamlessly integrating technology, environmental responsibility, and education, **GreenVolt redefines waste as a valuable resource, unlocking new possibilities for sustainability in schools and beyond.**

● Idea Overview:

GreenVolt is an integrated waste-to-energy system that leverages AI, automation, and heat recovery technologies to transform school sustainability:

AI-Powered Smart Bins: Use vision sensors, weight sensors, and infrared technology to classify and sort waste with high accuracy, improving recycling rates and minimizing contamination.

Composting & Heat Recovery: Organic waste is processed on-site, with the heat generated captured and stored for energy applications.

Waste Heat Utilization: Recovered heat from composting, electrical systems, and ambient sources is repurposed using **thermoelectric generators (TEGs), heat exchangers, and phase-change materials**, supplying energy for lighting, water heating, and other school needs.

This model **eliminates inefficiencies** in school waste management while introducing an **energy-saving element—a first-of-its-kind approach** in an educational setting.

● Key Participants:

Internal Team:

Developers & Researchers: Leading AI integration, composting, and energy storage solutions.



School Administration & Facilities Team: Assisting in implementation and logistics.

External Partners:

1. **Sparklo** – Implementing a **point-based reward system** for waste sorting.
2. **PowerBear** – Supplier of durable smart bin infrastructure.
3. **Tadweer (Abu Dhabi Waste Management Center)** – Partnering for large-scale waste optimization.

These stakeholders ensure GreenVolt is **technologically robust, cost-effective, and scalable**.

● **Costing:**

Our idea can **definitely be funded within the \$200,000 budget** allocated for the entire action plan, with a **contingency reserve** to account for unexpected costs. Details have been provided in the cost sheet below. To ensure **long-term sustainability beyond the initial funding**, GreenVolt will implement **low-maintenance designs, periodic system optimizations, and potential partnerships with local waste management and energy companies**. Additionally, the cost savings from **reduced waste disposal fees and energy efficiency improvements** will help offset operational expenses, keeping the system financially viable for years to come.

● **Scalability:**

GreenVolt's model can be adapted to other **schools, universities, and community spaces**, offering a scalable approach to sustainability. **Modular AI-powered bins and flexible heat recovery systems** allow customization based on waste volume and energy needs. While initial funding covers deployment in one school, **expansion to multiple institutions may require additional investment**. Partnering with **local waste management services, energy consultants, and educational bodies** can facilitate broader adoption, ensuring long-term impact in both **academic and community settings**.

● **Positive Outcomes Predicted:**

Environmental Impact

- **30-40% landfill waste reduction**
 - **Lower CO₂ emissions from waste decomposition**
 - **Improved recycling efficiency**
-



School & Community Impact

- **Reduced energy consumption** for heating and lighting
- **Engages students & staff in sustainability efforts**
- **Sets a model for future eco-friendly school infrastructure**

Long-Term Sustainability

- Encourages **green innovation** in education
- Can be adapted for **community & commercial use**
- Contributes to **UAE's Net Zero 2050 goal**

OPERATIONAL PLAN/ROADMAP/FINANCIALS

Sourcing and Fulfilment:

For the GreenVolt project, key components such as AI-powered smart bins, composting units, and thermal storage systems will be sourced from trusted suppliers specializing in sustainable technologies. We will work with local distributors for waste management technology and renewable energy components, ensuring we maintain a cost-effective and eco-friendly supply chain. We plan to store and manage key components in a central location within the school's facilities, with detailed tracking via inventory management software. This system will allow us to monitor stock levels and track the maintenance and replacement needs of critical parts.

- **Technology:**

Technology is at the core of GreenVolt's success. The AI-powered smart bins will use sensors and machine learning algorithms to sort waste efficiently, while composting units and thermal storage systems will transform waste into valuable energy. We will rely on cloud-based platforms to store and process data from these systems. Security measures will be implemented to protect user data, including encryption and secure access protocols. Regular data backups will be performed to ensure that in the case of any system failure, data can be restored and operations can continue without disruption. A solid disaster recovery plan will be in place to address potential technical issues promptly.

- **Community Impact:**

GreenVolt aims to have a lasting and far-reaching impact on the community. In the short term, we will monitor waste reduction, energy savings, and the engagement of students in sustainability initiatives.



Key performance indicators (KPIs) such as increased recycling rates, reduced energy costs, and student participation in environmental education programs will help measure the success of the project. In the long term, we will assess the wider community's adoption of sustainable practices, with feedback from students, families, and local businesses on how the initiative has influenced their habits. A point-based incentive system will track community engagement and reward active participation.

- **Key Team Members:**

Our team includes students with expertise in environmental science, technology, and project management. Key roles will include a project coordinator, a technical lead for AI and waste management systems, and a sustainability coordinator for outreach and education. We will also work closely with school administration, local environmental experts, and external suppliers to ensure smooth implementation. To maintain the project beyond the first year, we plan to establish a student-led sustainability committee responsible for ongoing monitoring, outreach, and expansion to other schools.

- **School Facilities:**

Yes, GreenVolt will require access to school facilities for the installation of smart bins, composting units, and thermal storage systems. The project will utilize existing school infrastructure for the placement of these units, ensuring minimal disruption to regular activities. We will also require space for tracking systems and the storage of equipment and materials. The project will enhance school facilities by creating a sustainable environment that can be leveraged for educational purposes, allowing students to directly engage with cutting-edge technologies that have real-world applications. After the initial phase, the impact on school facilities will include reduced waste management costs and energy savings, contributing to the school's long-term sustainability goals.

STRENGTHS

- Automated Sorting Accuracy
- Energy & Cost Savings
- Strong research background
- Carbon Footprint Reduction
- Potential grants and sponsorships
- Combines waste management and energy recovery
- Reduces Landfill Waste

OPPORTUNITIES

- Expand into universities, corporate campuses, and smart cities
- Aligns with circular economy initiatives



WEAKNESSES

- High initial setup cost
- Dependence on Participation
- Technical challenges
- Difficulty obtaining long-term financial backing
- Lack of awareness

THREATS

- Regulatory Changes



GREENVOLT

- Contributes to scientific research
- Educational & Corporate Recognition
- Collaborate with waste management firms



ROADMAP Cont... FINANCIAL PLAN

Creating a financial plan is where all the business planning comes together. This section will help you estimate your expenditure and pinpoint key milestones for funding release. Carefully identifying expected expenditure to ensure your project is a success.

At a minimum, this section should include a start-up cost and projected expenditure for the first year, alongside a summary of assumptions for how you are making these projections.

Here is how you set up your financial plan:

- **Projected Start-Up Costs:** What do you need to spend to get started? The table below provides a template for recording predicted expenditure. It is important to estimate the cash required to reach certain milestones for the project. This will be used as a metric to assess the success and therefore future release of funding.
- **First Year Costs:** The table allows you to project the expenditure required in the first year of the project. The project will be reviewed at the end of each term and funding will be released accordingly. This is an estimate and may change as the project progresses.



START – COSTS: FIRST INSTALLMENT

COST ITEMS	DATE DUE	BUDGET	ACTUAL	UNDER/OVER
Licences/Registration				
Permits (Municipality, Chamber of Commerce, MoIAT, MOHRE, ADDC)	N/A			
Insurance	N/A			
Legal (Approval & Compliance)	N/A			
Consultation	N/A			
Training				
Software (General)				
Miscellaneous				
SCHOOL				
Space Rental/Lease				
Utility Costs	N/A			
Furniture	N/A	N/A	N/A	N/A
Equipment				
Hardware				
Software (Monitoring system - ThingSpeak)				
Installation Fees				
Start up Resources				
Miscellaneous				
LABOUR EXPENSES				
Training				
Miscellaneous				
OTHER				
Miscellaneous				
ESTIMATED START-UP BUDGET				

Project Milestones FIRST YEAR

cOSTS	TERM 1 (INCLUDE sTART UP)	TERM 2	TERM 3	FIRST YEAR COSTS
START-UP COST				
TOTAL EXPENDITURE				

ASSESSMENT AND MONITORING PLAN

Initial Start-Up of Project

Months 1-3: Infrastructure Setup & Foundation Development

- Deploy **AI-powered smart bins** in key locations, integrating **vision sensors, weight sensors, and infrared technology** for accurate waste classification.
- Install **on-site composting units** and **heat recovery systems** to begin initial waste-to-energy conversion.
- Establish a **basic database** where sensor data will be stored and analyzed, providing foundational insights into waste patterns.
- While starting with a simple data collection system, the ThingSpeak monitoring system, we plan to **expand to a centralized monitoring dashboard or mobile app** as funding allows, enabling real-time tracking, predictive analytics, and user engagement.

Months 4-6: Testing, Calibration & Pilot Implementation

- Conduct thorough system testing and calibration to ensure accuracy in waste sorting, composting, and heat capture.
- Launch a **pilot program** within designated school areas to test system performance and identify any required adjustments.
- Engage students and staff in **workshops and training** to ensure proper waste disposal practices and encourage system use.

Months 7-9: Full-Scale Deployment & Operation

- Begin **full-scale deployment** of the GreenVolt system, adding on to the bins, across the school campus, expanding waste sorting, composting, and heat recovery operations to all relevant areas.
- Monitor system performance in real-time and continue **data collection** for further analysis and optimization.

Months 10-12: Performance Optimization & Efficiency Improvements

- Use the collected data to **optimize system performance**, identifying opportunities for improved waste sorting accuracy, composting efficiency, and energy recovery.
- Implement necessary adjustments and enhancements to improve system efficiency based on insights gathered during the first nine months.

Community Impact

GreenVolt will engage students, families, and the local community by actively involving them in sustainability practices. Students will gain practical experience with AI and waste management systems, fostering environmental awareness. The project's real-time tracking of waste management efforts will encourage families to adopt sustainable practices, extending the impact of GreenVolt beyond the school environment.

Our monitoring system: The **ThingSpeak monitoring system** works by collecting sensor data from the auto-sorting bins and sending it to the cloud for analysis. Here's how it functions:

1. Data Collection (Sensors in the Bins)

Each **auto-sorting bin** is equipped with multiple sensors to monitor waste disposal:

- **Weight Sensors (Honeywell Model 13 Series):** Measure the weight of the waste.
- **Vision Sensors (Keyence IV Series):** Identify waste type (plastic, paper, organic, etc.).
- **Infrared Sensors (HC-SR501 PIR):** Detect motion when someone approaches or disposes of waste.

2. Data Transmission (IoT Connectivity)

- The sensors in each bin **collect real-time data** and send it to a **microcontroller** (e.g., Arduino, Raspberry Pi, or ESP8266).
- The microcontroller processes the data and transmits it to **ThingSpeak** via Wi-Fi or a cellular network.

3. Cloud Processing (ThingSpeak Platform)

- ThingSpeak **receives the data** and stores it in the cloud.
- It **visualizes data** using graphs and dashboards, helping users analyze waste disposal trends.
- It can also trigger **alerts** when bins are full or when abnormal waste patterns are detected.

4. Action & Decision Making

- The school or waste management team can **monitor bin levels remotely** and decide when to collect waste.
- The system can also help **optimize waste collection schedules**, reducing unnecessary pickups.
- It can **predict trends** in waste generation to improve recycling and sustainability efforts.

This cost covers the **ThingSpeak cloud service subscription** and **data transmission costs** for remote monitoring.

Social Impact

GreenVolt will empower students by providing leadership opportunities through active participation in sustainability initiatives. Students will take charge of waste sorting, energy-saving practices, and championing environmental causes within the school community. By integrating a point-based incentive system, students will be encouraged to lead by example, earning rewards for their contributions to waste management and energy conservation efforts. This system will foster a sense of responsibility, teamwork, and accountability, allowing students to develop valuable leadership skills while driving a collective movement towards sustainability.

More info on the point based system will be given at the end of the document.

Physical Impact

GreenVolt will enhance school infrastructure by repurposing organic waste and capturing waste heat, reducing reliance on external energy sources and lowering emissions. The system improves waste management efficiency and helps the school operate more sustainably.

Key Metrics:

- **30-40% reduction in waste-related CO₂ emissions:** Reduced methane emissions and energy consumption from waste heat recovery.
- **25% improvement in waste disposal efficiency:** AI-powered bins and on-site composting reduce waste sent to landfills.
- **10-15% reduction in energy costs:** Heat recovery decreases the need for external energy, lowering bills.

Positive Environmental & Climate Behavioural Change

GreenVolt fosters a transformative shift in sustainability by making waste management and energy conservation interactive, engaging, and data-driven. Real-time monitoring and the point-based incentive system not only offer insight into waste reduction and energy savings but also motivate students and staff to take action.

A study from a school in the UK that introduced waste sorting and energy-saving measures reported a 45% increase in recycling rates and a 30% decrease in energy consumption over the first year.

Similarly, schools in Canada that integrated composting and waste heat recovery saw a 40% reduction in landfill waste, with students engaging in sustainability initiatives more regularly.

This model promotes long-term behavioral change, empowering the entire school community to adopt eco-friendly habits and become champions of environmental responsibility.

Scalability

GreenVolt is purposefully designed for scalability, offering a flexible solution that can be adapted to various school and community settings. The system's minimal infrastructure requirements and low operational costs make it feasible for expansion. Partnerships will play a key role in ensuring its successful rollout.

Initially, we plan to collaborate with nearby schools and local businesses, such as cafés and grocery stores, to integrate their organic waste into our system. With the necessary legal permissions, designated bins will be set up at strategic locations for efficient waste deposit, allowing more organic material to be processed and converted into heat, enhancing the system's energy output.

This approach not only supports local waste reduction efforts but also fosters a sense of shared responsibility, with the community contributing to the sustainability goals of the school. By leveraging these partnerships, we anticipate a broadening of GreenVolt's impact beyond a single institution, enabling replication in other schools and expanding to community centers, local governments, and even businesses that want to reduce their environmental footprint. This scalable model ensures that the project can grow, benefiting multiple stakeholders while maintaining the same efficiency and low-cost operations.

Additional Information

1. Competitor Analysis

Company	Product	Similarity to Our Project	Approximate Cost	Limitations	Competitive Edge of Our Project	Market Share
TOMRA Recycling Sorting	Sensor-based sorting technology for high-purity recycle recovery.	Automated sorting for better waste management.	\$300,000 - \$500,000 per system	High initial investment; primarily designed for large-scale facilities, not easily adaptable to smaller setups.	More cost-efficient through the point-based incentive system to encourage community participation.	Global market leader in recycling technology; significant market share in automated sorting globally.
Envac Dubai LLC	Pneumatic waste collection systems.	Automated waste handling to streamline collection.	\$500,000 - \$1,000,000 per system	Expensive setup and maintenance; limited flexibility in integrating sorting technologies.	More focused on efficient sorting with a scalable solution tailored to UAE's waste characteristics.	Significant presence in the UAE, but limited global reach compared to competitors like TOMRA.
Imdaad	Integrated waste management solutions, including recycling services.	Commitment to sustainability and waste reduction.	\$200,000 - \$400,000 per system	Less focus on automation and more on general waste management, making it less specialized for sorting needs.	Emphasis on community-driven recycling with incentives, improving both recycling rates and cost-effectiveness.	Strong local presence in the UAE, but limited international market share.
Bee'ah	Advanced recycling facilities with a mix of manual, mechanical, and automated sorting.	Use of automated sorting to improve efficiency.	\$500,000 - \$750,000 per system	High operational costs; reliant on a combination of manual and automated processes, reducing overall efficiency.	Integration of point system to increase social impact and community involvement, enhancing sustainability.	Leading waste management company in the UAE with a growing presence in the region.
Veolia Environmental Services	Comprehensive waste management and recycling services.	Focus on sustainable waste management practices.	\$300,000 - \$600,000 per system	Generic solutions that may not fully meet the specific needs of the UAE's waste management challenges.	Designed for local adaptation with a focus on community participation and efficient waste sorting.	Global presence with a substantial market share in waste management, particularly in Europe and Asia.

Here we've included a competitor analysis of national and international companies using similar mechanisms. The data highlights that our project outperforms others in key areas such as efficiency, cost-effectiveness, and scalability.

2. Detailed Carbon Offset Calculations

1. Carbon Footprint

Auto-Sorting Bins

- **Manufacturing (Sensors, Bins, Electronics):**
Footprint: ~70 kg CO₂ per bin.
Total Emissions (5 bins): $70 \times 5 = 350 \text{ kg CO}_2$.
- **Power for Sensors:**
Assuming 0.5 kWh/day/bin, powered by solar: **0 kg CO₂** operational emissions.

Composting Unit

- **Manufacturing and Installation:**
Total Emissions: 1 unit $\times$ 250 = **250 kg CO₂**.
- **Methane Emissions (If Poorly Managed):**
Proper management reduces anaerobic zones to <5%.
Annual Methane Emissions: $5 \times 0.05 \times 365 = 91 \text{ kg CO}_2/\text{year}$.

Thermal Tank and Stirling Engine

- **Manufacturing:**
Thermal tank and pipes: **300 kg CO₂**.
Stirling engine: **400 kg CO₂**.
Total: $300 + 400 = 700 \text{ kg CO}_2$.

Tracking Program

- **Cloud Emissions:**
Cloud storage: 0.5 kg CO₂/GB/month.
Total Emissions (10 GB/year): $0.5 \times 10 = 5 \text{ kg CO}_2/\text{year}$.

Total Carbon Footprint (Year 1):

- Auto-sorting bins: **350 kg CO₂**
- Composting unit: **341 kg CO₂** (including methane emissions).
- Thermal tank and Stirling engine: **700 kg CO₂**
- Tracking program: **5 kg CO₂**
- **Total: ~1,396 kg CO₂**

2. Carbon Offset

Methane Avoidance (Landfill Diversion)

- Methane from 1 ton of organic waste in a landfill: ~1,500 kg CO₂ equivalent.
- Organic waste diverted: 2 kg/day = 730 kg/year = 0.73 tons/year.
- Avoided Methane Emissions: $0.73 \times 1,500 = 1,095 \text{ kg CO}_2/\text{year}$.

Compost Use in Agriculture

- Replacing chemical fertilizers reduces emissions (~5 kg CO₂ per kg of fertilizer).
- Compost produced: ~200 kg/year.
- Offset: $200 \times 5 = 1,000 \text{ kg CO}_2/\text{year}$.

Electricity Generation (Stirling Engine)

- Energy generated: ~2 kWh/day.
- Grid electricity emissions: ~0.5 kg CO₂/kWh.
- Offset: $2 \times 365 \times 0.5 = 365 \text{ kg CO}_2/\text{year}$.

Total Carbon Offset (Year 1):

- Methane avoidance: **1,095 kg CO₂**
- Compost use: **1,000 kg CO₂**
- Electricity generation: **365 kg CO₂**
- **Total: ~2,460 kg CO₂**

Net Carbon Impact

- Offset: **2,460 kg CO₂**
- Footprint: **1,396 kg CO₂**
- **Net Offset: ~1,064 kg CO₂ positive (Year 1)**

Conclusion

The system achieves a significant carbon offset, exceeding its footprint by ~76%. With proper management and scaling, this system can remain carbon-positive and contribute to environmental sustainability.

3. Sparklo - Point based reward system

Sparklo Partnership

For our autosorting bins, we have partnered with Sparklo, a company known for its Sparklo Recycling RVM (Reverse Vending Machine). Instead of using their standard machine, we will be customizing it into a bin concept while maintaining their successful point-based reward system. However, we will not be including the reward system in our current cost estimates, as we plan to integrate it only after ensuring the smooth operation of the system in its first year.

Sparklo's reward system has already proven to be highly effective in GEMS Cambridge International Abu Dhabi (CIA), where it has encouraged students to recycle more by offering incentives for depositing plastic bottles. Seeing its success in a fellow Cambridge school, we aim to implement a similar approach to drive engagement and make recycling a habit in our own school community.

About Sparklo

Sparklo specializes in smart recycling solutions through its RVM technology, which allows users to deposit plastic bottles in exchange for points redeemable for rewards. Their machines are designed to be user-friendly, scalable, and easily adaptable to different environments, from schools to corporate offices. By integrating their technology into our autosorting bins, we aim to enhance recycling efficiency while also creating a long-term behavioral change in waste management.

The monthly leasing fee for the Sparklo RVM is AED 1,745 for the indoor model and AED 2,020 for the outdoor model, covering all associated services, including:

- ✓ Design customization of the RVM
- ✓ Delivery & installation
- ✓ Full-year maintenance
- ✓ Unlimited access to Sparklo rewards, including discounts at Carrefour, Lulu, ADNOC, and more.